

Calculus II

Rational numbers to decimal notation

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Algorithm (Write a fraction to periodic base 10)

Input: a fraction $\frac{p}{q}$. Output: integer A and two groups of digits such that $\frac{p}{q} = A + 0.b_1 \dots b_m \overline{c_1 \dots c_n}$.

- 1 Initialize **digits** = () as the empty sequence.
- 2 Initialize **remainders** = () as the empty sequence.
- 2 Divide p by q with remainder r and set A to be the quotient.
Append r to remainders.
- 3 While r is not equal to zero:
 - 3.1 Multiply r by 10.
 - 3.2 Divide the result by q with remainder r' and quotient d .
 - 3.3 If r' belongs to **remainders** with first occurrence at position $m + 1$, slice digits into two sequences $b_1, \dots, b_m$ and $c_1, \dots, c_n$. Return A and $b_1, \dots, b_m, c_1, \dots, c_n$ as the desired digits.
 - 3.4 Append d to **digits** and r to **remainders**.
 - 3.5 Set $r = r'$ and go back to Step 3.
- 4 If r attained the value 0 in the execution of the loop, the fraction $\frac{p}{q}$ has a finite decimal representation given by A and **digits**.

Algorithm (Write a fraction to periodic base X)

Input: a fraction $\frac{p}{q}$. Output: integer A and two groups of digits such that $\frac{p}{q} = A + 0.b_1 \dots b_m \overline{c_1 \dots c_n}$.

- 1 Initialize **digits** = () as the empty sequence.
- 2 Initialize **remainders** = () as the empty sequence.
- 2 Divide p by q with remainder r and set A to be the quotient.
Append r to remainders.
- 3 While r is not equal to zero:
 - 3.1 Multiply r by X.
 - 3.2 Divide the result by q with remainder r' and quotient d .
 - 3.3 If r' belongs to **remainders** with first occurrence at position $m + 1$, slice digits into two sequences $b_1, \dots, b_m$ and $c_1, \dots, c_n$. Return A and $b_1, \dots, b_m, c_1, \dots, c_n$ as the desired digits.
 - 3.4 Append d to **digits** and r to **remainders**.
 - 3.5 Set $r = r'$ and go back to Step 3.
- 4 If r attained the value 0 in the execution of the loop, the fraction $\frac{p}{q}$ has a finite decimal representation given by A and **digits**.

Example

Convert $\frac{86}{7}$ to repeating decimal notation.

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder.

$$86 = ? \cdot 7 + ? \quad |$$

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$$86 = 12 \cdot 7 + 2 \quad |$$

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Example

Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$86 = 12 \cdot 7 + 2 \quad \Bigg| \quad \frac{86}{7} = 12 + \dots$$

2

Example

Convert $\frac{86}{7}$ to repeating **decimal notation**.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$86 = 12 \cdot 7 + 2 \quad \Bigg| \quad \frac{86}{7} = 12 + \dots$$

$$2 \cdot 10 = 20$$

Example

Convert $\frac{86}{7}$ to repeating **decimal notation**.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$2 \cdot 10 = 20 \qquad 86 = 12 \cdot 7 + 2 \quad \Bigg| \quad \frac{86}{7} = 12 + \dots$$

Example

Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$\begin{array}{rcll}
 & 86 & = & 12 \cdot 7 + 2 \\
 2 \cdot 10 & = & \color{red}{20} & = \color{red}{?} \cdot 7 + \color{red}{?}
 \end{array}
 \quad \Bigg| \quad
 \frac{86}{7} = 12 + \dots$$

Example

Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$\begin{array}{rclcl}
 & 86 & = & 12 \cdot 7 + 2 & \Bigg| & \frac{86}{7} & = & 12 + \dots \\
 2 \cdot 10 & = & \textcolor{red}{20} & = & \textcolor{red}{2} \cdot 7 + \textcolor{red}{6} & & & &
 \end{array}$$

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$\begin{array}{rclcl}
 & 86 & = & 12 \cdot 7 + 2 & \left| \frac{86}{7} = 12 + \dots \right. \\
 2 \cdot 10 & = & 20 & = & \textcolor{red}{2} \cdot 7 + 6 & \left| \phantom{\frac{86}{7}} = 12.\textcolor{red}{2} + \dots \right.
 \end{array}$$

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$\begin{array}{rclcl}
 & 86 & = & 12 \cdot 7 + 2 & \left| \frac{86}{7} = 12 + \dots \right. \\
 2 \cdot 10 & = & 20 & = & 2 \cdot 7 + \textcolor{red}{6} & = 12.2 + \dots \\
 \textcolor{red}{6} \cdot 10 & = & 60 & & &
 \end{array}$$

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$\begin{array}{rclcl}
 & 86 & = & 12 \cdot 7 + 2 & \left| \frac{86}{7} = 12 + \dots \right. \\
 2 \cdot 10 & = & 20 & = & 2 \cdot 7 + 6 & = 12.2 + \dots \\
 6 \cdot 10 & = & 60 & = & ? \cdot 7 + ? &
 \end{array}$$

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Divide with 86 by 7 with remainder. Repeat using remainder ...

$$\begin{array}{rclcl}
 & 86 & = & 12 \cdot 7 + 2 & \left| \frac{86}{7} = 12 + \dots \right. \\
 2 \cdot 10 & = & 20 & = & 2 \cdot 7 + 6 & = 12.2 + \dots \\
 6 \cdot 10 & = & \textcolor{red}{60} & = & \textcolor{red}{8} \cdot \textcolor{red}{7} + \textcolor{red}{4} & = 12.28 + \dots
 \end{array}$$

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

$$\begin{array}{rclcl}
 & 86 & = & 12 \cdot 7 + 2 & \left| \frac{86}{7} = 12 + \dots \right. \\
 2 \cdot 10 & = & 20 & = & 2 \cdot 7 + 6 & = 12.2 + \dots \\
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 & 86 & = & 12 \cdot 7 + 2 & \left| \frac{86}{7} = 12 + \dots \right. \\
 2 \cdot 10 & = & 20 & = & 2 \cdot 7 + 6 & = 12.2 + \dots \\
 6 \cdot 10 & = & 60 & = & 8 \cdot 7 + 4 & = 12.28 + \dots \\
 \textcolor{red}{4} \cdot 10 & = & 40 & & &
 \end{array}$$

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Divide with 86 by 7 with remainder. Repeat using remainder ...

		$86 = 12 \cdot 7 + 2$	$\left \frac{86}{7} = 12 + \dots$
$2 \cdot 10$	$= 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
$6 \cdot 10$	$= 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
$4 \cdot 10$	$= 40$	$= ? \cdot 7 + ?$	

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Divide with 86 by 7 with remainder. Repeat using remainder ...

$$\begin{array}{rclcl}
 & 86 & = & 12 \cdot 7 + 2 & \left| \frac{86}{7} = 12 + \dots \right. \\
 2 \cdot 10 & = & 20 & = & 2 \cdot 7 + 6 & = 12.2 + \dots \\
 6 \cdot 10 & = & 60 & = & 8 \cdot 7 + 4 & = 12.28 + \dots \\
 4 \cdot 10 & = & \color{red}{40} & = & \color{red}{5 \cdot 7 + 5} & = 12.285 + \dots
 \end{array}$$

Example

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Divide with 86 by 7 with remainder. Repeat using remainder ...

		$86 = 12 \cdot 7 + 2$	$\left \frac{86}{7} = 12 + \dots$
$2 \cdot 10 = 20$	$= 2 \cdot 7 + 6$		$= 12.2 + \dots$
$6 \cdot 10 = 60$	$= 8 \cdot 7 + 4$		$= 12.28 + \dots$
$4 \cdot 10 = 40$	$= 5 \cdot 7 + 5$		$= 12.285 + \dots$
$5 \cdot 10 = 50$			

Example

Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

		$86 = 12 \cdot 7 + 2$	$\left \frac{86}{7} = 12 + \dots$
$2 \cdot 10$	$= 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
$6 \cdot 10$	$= 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
$4 \cdot 10$	$= 40$	$= 5 \cdot 7 + 5$	$= 12.285 + \dots$
$5 \cdot 10$	$= 50$	$= ? \cdot 7 + ?$	

Example

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Divide with 86 by 7 with remainder. Repeat using remainder ...

	$86 = 12 \cdot 7 + 2$	$\left \frac{86}{7} = 12 + \dots$
$2 \cdot 10 = 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
$6 \cdot 10 = 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
$4 \cdot 10 = 40$	$= 5 \cdot 7 + 5$	$= 12.285 + \dots$
$5 \cdot 10 = 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$

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Divide with 86 by 7 with remainder. Repeat using remainder ...

		$86 = 12 \cdot 7 + 2$	$\left \frac{86}{7} = 12 + \dots$
$2 \cdot 10$	$= 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
$6 \cdot 10$	$= 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
$4 \cdot 10$	$= 40$	$= 5 \cdot 7 + 5$	$= 12.285 + \dots$
$5 \cdot 10$	$= 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10$	$= 10$		

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Divide with 86 by 7 with remainder. Repeat using remainder ...

		$86 = 12 \cdot 7 + 2$	$\left \frac{86}{7} = 12 + \dots$
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$6 \cdot 10$	$= 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
$4 \cdot 10$	$= 40$	$= 5 \cdot 7 + 5$	$= 12.285 + \dots$
$5 \cdot 10$	$= 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10$	$= 10$	$= ? \cdot 7 + ?$	

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

	$86 = 12 \cdot 7 + 2$	$\frac{86}{7} = 12 + \dots$
$2 \cdot 10 = 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
$6 \cdot 10 = 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
$4 \cdot 10 = 40$	$= 5 \cdot 7 + 5$	$= 12.285 + \dots$
$5 \cdot 10 = 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10 = 10$	$= 1 \cdot 7 + 3$	$= 12.28571 + \dots$

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

		$86 = 12 \cdot 7 + 2$	$\frac{86}{7} = 12 + \dots$
$2 \cdot 10$	$= 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
$6 \cdot 10$	$= 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
$4 \cdot 10$	$= 40$	$= 5 \cdot 7 + 5$	$= 12.285 + \dots$
$5 \cdot 10$	$= 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10$	$= 10$	$= 1 \cdot 7 + 3$	$= 12.28571 + \dots$
$3 \cdot 10$	$= 30$		

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Divide with 86 by 7 with remainder. Repeat using remainder ...

		$86 = 12 \cdot 7 + 2$	$\left \frac{86}{7} = 12 + \dots$
$2 \cdot 10$	$= 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
$6 \cdot 10$	$= 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
$4 \cdot 10$	$= 40$	$= 5 \cdot 7 + 5$	$= 12.285 + \dots$
$5 \cdot 10$	$= 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10$	$= 10$	$= 1 \cdot 7 + 3$	$= 12.28571 + \dots$
$3 \cdot 10$	$= 30$	$= ? \cdot 7 + ?$	

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Divide with 86 by 7 with remainder. Repeat using remainder ...

	$86 = 12 \cdot 7 + 2$	$\frac{86}{7} = 12 + \dots$
$2 \cdot 10 = 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
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$5 \cdot 10 = 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10 = 10$	$= 1 \cdot 7 + 3$	$= 12.28571 + \dots$
$3 \cdot 10 = 30$	$= 4 \cdot 7 + 2$	$= 12.285714 + \dots$

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Divide with 86 by 7 with remainder. Repeat using remainder ...

	$86 = 12 \cdot 7 + 2$	$\frac{86}{7} = 12 + \dots$
$2 \cdot 10 = 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
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$5 \cdot 10 = 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10 = 10$	$= 1 \cdot 7 + 3$	$= 12.28571 + \dots$
$3 \cdot 10 = 30$	$= 4 \cdot 7 + 2$	$= 12.285714 + \dots$
$2 \cdot 10 = 20$		

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$5 \cdot 10 = 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10 = 10$	$= 1 \cdot 7 + 3$	$= 12.28571 + \dots$
$3 \cdot 10 = 30$	$= 4 \cdot 7 + 2$	$= 12.285714 + \dots$
$2 \cdot 10 = 20$	$= \dots$	$= 12.\overline{285714} + \dots$

... until the remainder repeats.

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

	$86 = 12 \cdot 7 + 2$	$\frac{86}{7} = 12 + \dots$
$2 \cdot 10 = 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
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$4 \cdot 10 = 40$	$= 5 \cdot 7 + 5$	$= 12.285 + \dots$
$5 \cdot 10 = 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10 = 10$	$= 1 \cdot 7 + 3$	$= 12.28571 + \dots$
$3 \cdot 10 = 30$	$= 4 \cdot 7 + 2$	$= 12.285714 + \dots$
$2 \cdot 10 = 20$	$= \dots$	$= 12.\overline{285714} + \dots$

... until the remainder repeats.

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Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

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$3 \cdot 10 = 30$	$= 4 \cdot 7 + 2$	$= 12.285714 + \dots$
$2 \cdot 10 = 20$	$= \dots$	$= 12.\overline{285714} + \dots$

... until the remainder repeats. Answer:

$$\frac{86}{7} = 12.\overline{285714}.$$

Example

Convert $\frac{86}{7}$ to repeating decimal notation.

Divide with 86 by 7 with remainder. Repeat using remainder ...

	$86 = 12 \cdot 7 + 2$	$\frac{86}{7} = 12 + \dots$
$2 \cdot 10 = 20$	$= 2 \cdot 7 + 6$	$= 12.2 + \dots$
$6 \cdot 10 = 60$	$= 8 \cdot 7 + 4$	$= 12.28 + \dots$
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$5 \cdot 10 = 50$	$= 7 \cdot 7 + 1$	$= 12.2851 + \dots$
$1 \cdot 10 = 10$	$= 1 \cdot 7 + 3$	$= 12.28571 + \dots$
$3 \cdot 10 = 30$	$= 4 \cdot 7 + 2$	$= 12.285714 + \dots$
$2 \cdot 10 = 20$	$= \dots$	$= 12.\overline{285714} + \dots$

... until the remainder repeats. Answer:

$$\frac{86}{7} = 12.\overline{285714}.$$

Example

Convert $\frac{2}{13}$ to repeating decimal notation.

	2	=	$0 \cdot 13 + 2$	$\frac{2}{13}$	=	$0 + \dots$
$2 \cdot 10$	=	20	=	$1 \cdot 13 + 7$	=	$0.1 + \dots$
$7 \cdot 10$	=	70	=	$5 \cdot 13 + 5$	=	$0.15 + \dots$
$5 \cdot 10$	=	50	=	$3 \cdot 13 + 11$	=	$0.153 + \dots$
$11 \cdot 10$	=	110	=	$8 \cdot 13 + 6$	=	$0.1538 + \dots$
$6 \cdot 10$	=	60	=	$4 \cdot 13 + 8$	=	$0.15384 + \dots$
$8 \cdot 10$	=	80	=	$6 \cdot 13 + 2$	=	$0.153846 + \dots$
$2 \cdot 10$	=	20	=	$1 \cdot 13 + 7$	=	$0.1538461 + \dots$
...					=	$0.\overline{153846} + \dots$

Answer: $\frac{2}{13} = 0.\overline{153846}$